

Power Quality Improvement in Solar Powered Standalone System using D-STATCOM

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Abstract— As there is a huge growth in the renewable energy sources, the solar farm based power plants are getting more popular among all researchers. In the large power systems, the sudden variations in the load requirements are more common. The variations in the load lowers the power quality in the power system. This paper concentrates on the power quality improvement in 100kW solar based standalone power system. In order to improve the power quality in the large power systems, the FACTS devices such as Static Synchronous Series Compensator (SSSC), Static Shunt Compensator (STATCOM), Unified Power Flow Controller (UPFC) etc., are used. Here, the STATCOM is implemented to improve the voltage profile of the power system under varying load conditions. The Distribution Static Compensator (D-STATCOM) balances the deficiency of reactive power by maintaining the voltage profile in the system.

Keywords— *Quadratic DC-DC Buck boost Converter, PV Panel, Cascaded Converter, Conventional Buck-Boost Converter, MPPT*

I. INTRODUCTION

With the growing need of clean and green energy, there has been a major shift from generating electricity through fossil fuels to opting renewable sources for energy generation. As of March 2020, 35.86% of India's installed electric capacity contributes to energy from renewable sources. Hence the development of solar panel based power plant has become more popular nowadays. As the solar radiations changes time to time, there is an unregulated output across the panel. For the regulation of the output voltage, the solar energy system requires DC-DC converter operated with MPPT algorithm [1-4]. The MPPT controller is mandate for achieving maximum power from the solar panel [5-9]. At the consumer end, almost all the loads are operated with AC supply. As the solar panel outputs DC, for operating AC loads, the DC to AC conversion is mandatory. Hence the solar based system requires three phase inverter for converting DC from solar panel to AC. The common concerns at the load is, the load will not be constant always. The active power and reactive power requirements of the load varies time to time. These large variations in the power leads to poor power quality and poor power factor. This leads to high payment of electricity rate. Hence, there need to be an integration of FACTS devices such as SVC, STATCOM, SSSC, etc., into the power system. The STATCOM is a shunt compensator used for the injection of the reactive power for the power quality and power factor improvement. It is the voltage source converter with the control. Traditionally, the controllers comprise of conventional PI controller with hysteresis current control scheme [10]. There are many existing reports on solving power quality issues using FACTS devices, esp. by STATCOM. The paper [11] explains the STATCOM based

hybrid grid connected power system comprising wind and a solar power source. The STATCOM is used to maintain the flat voltage profile at the point of common coupling and to improve the power quality. There are many control algorithms developed for the DSTSATCOM such as SRF, IRPT [12]. [13] proposes the comparison of two control theories of STATCOM such as IRPT and dq theory. The other control strategies like fuzzy based [14], ANN based [15], Neuro Fuzzy [16-17] based STATCOM controls are proposed.

This paper proposes power quality control of the solar based AC transmission system. The design of solar fed boost converter with MPPT and DSTATCOM with controller are discussed. The IRPT algorithm is used for the proposed system for the regulation of power system under non-linear load conditions and to maintain the flat voltage profile.

The system comprises of many subsystems such as solar array, boost converter with MPPT, three phase inverter, DSTATCOM, three phase filter and the nonlinear loads. The block diagram of the entire system is shown in Fig.1.

The section 2 describes the specifications of solar panel, design and MPPT control of boost converter, the three phase inverter control and the loads connected to the system. The section 3 discusses on design of STATCOM and its control. Section 4 explains the simulation analysis and the results. Section 5 completes the paper.

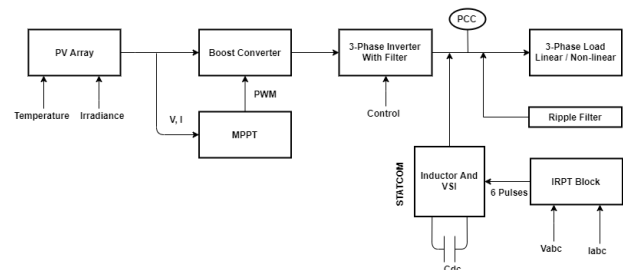


Fig.1 Block Diagram of the model

II. PV SYSTEM

A. Solar panel Specifications and its design

The PV panel characteristics of the of the 1 soltech 1STH-215-P,100kW solar panel is shown in Fig.2. The Solar PV array is designed with 10 series and 47 parallel strings. The design parameters of solar PV array is as follows: Maximum power is 100 kW and open circuit voltage is 363

V and max power Voltage is 290 V. The minimum DC voltage to Inverter to obtain 415 V three phase output is Minimum $V_{dc} = 1.2 \times 415 \times \sqrt{2} = 704.28$ V

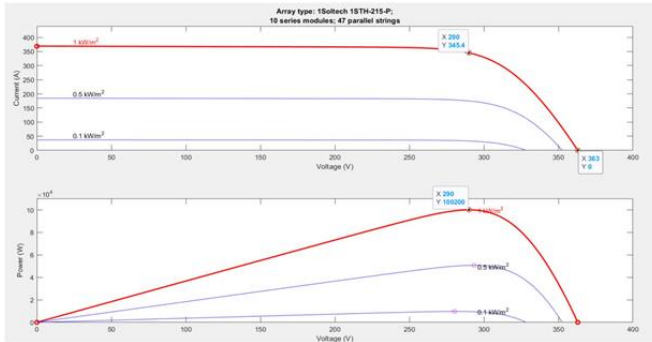


Fig. 2 PV and VI characteristics of solar array

B. Boost converter with MPPT –Inverter with load module

MPPT concept is used to generate the PWM required for the Boost Converter. MPPT controller is used to extract the maximum power from the PV module. Most of the commonly used MPPT algorithms are based on the concept of slope-tracking. Here, Perturb and Observe algorithm shown in Fig 3 is used. There are two factors such as duty ration and the magnitude of the perturbation voltage that affect the performance of P&O algorithms.

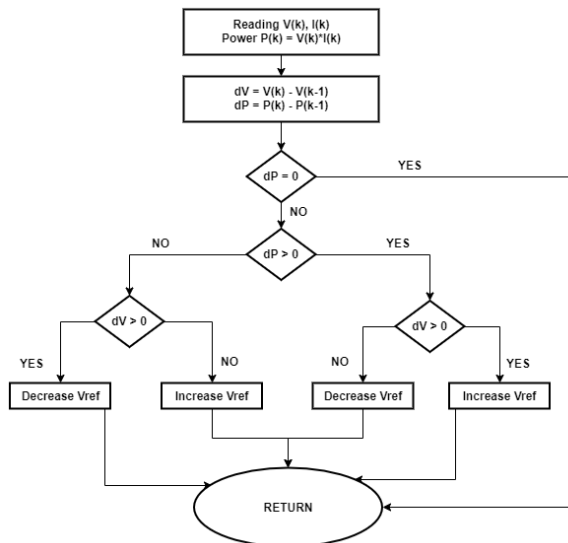


Fig. 3 P and O algorithm-MPPT

The Boost converter design should enable 700 V at the output terminals. Range of the solar input voltage is between 250 to 350 V. Total power of the system is 100 kW. Switching frequency is 5 kHz. The L and C values are 1.6 mH and 2626 μ F with the assumption of current ripple as 5% of input current and output voltage ripple as 1% of output voltage. The three phase inverter is tied after the boost converter to convert DC voltage into AC. The Sine wave PWM technique is used for the generation of the pulses to the inverter circuit. To make the three phase AC waveform smoother, the LC filter is kept at the output of the inverter [18].

The linear and non-linear loads are connected to this system, the resistive load of 10kW and the inductive load of 10kVAR are connected in the system. Similarly, the nonlinear load The specifications linear and nonlinear loads, the resistive and inductive loads of 10kW and 10kVAR are connected through the diode bridge. The shunt compensator D-STATCOM is used for the injection of reactive current. The design of STATCOM is discussed in the next section.

III. STATCOM MODELDESIGN

A. STATCOM Design

The DSTATCOM is the voltage Source converter and is used for the injection of reactive power. The basic circuit schematic of D-STATCOM is shown in Fig.4 [19].The design of STATCOM is given from the Eqns. [1] to [3].

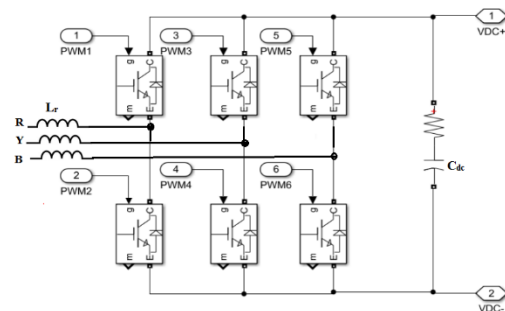


Fig. 4 Schematic of STATCOM

The line voltage of the D-STATCOM, $(V_{LL}) = 415$ V
DC voltage of the STATCOM

$$(V_{dc}) = 2 \times \sqrt{2} \times V_{LL} / m \times \sqrt{3} = 677.7 \text{ V} \approx 700 \text{ V} \quad (1)$$

Where the modulation index (m) is 0.8

$$\text{Value of series inductor } L_r = \frac{\sqrt{3} m V_{dc}}{12 a f_{sw} I_{ripple}} \quad (2)$$

Then the energy transfer is in time 30 ms is

$$\frac{1}{2} C_{dc} [V_{dc}^2 - V_{dc1}^2] = 3k \text{ Valt} \quad (3)$$

Where a is overload factor which is equal to 1.2.

Variation of energy k = 0.1

$V_{dc} = 700$ V,

5 to 8% dip in V_{dc} is V_{dc1} which is equal to 675 V

I (Source current) of the D-STATCOM is ranging between 100 A to 140 A

Current ripple through the inductor $L_r = I_{ripple} = 5\%$

$L_r \approx 4$ mH

DC capacitor of the STATCOM $C_{dc} = 12882.75 \mu$ F

$C_{dc} \approx 13000 \mu$ F

B. Control of STATCOM

IRPT Algorithm is used for the control of STATCOM. The block diagram of IRPT algorithm is shown below in Fig 5. This is used to find the reference current that has to be injected into the system through voltage source inverter.

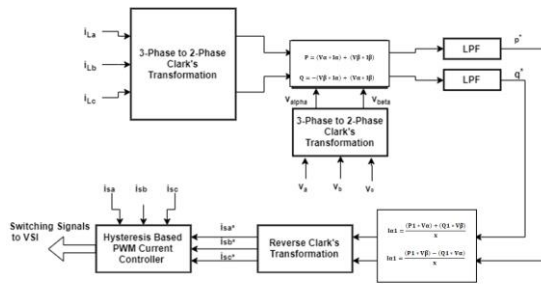


Fig. 5 Basic IRPT control algorithm

IV SIMULATION ANALYSIS

The schematic in Fig 6 displays the overall SIMULINK model of the proposed work. The solar PV array, the boost converter and the closed loop PI controller followed by the single phase inverter and LC filter. All the modules are discussed in the previous sections and the simulation is made to run and the resultant graphs are documented.

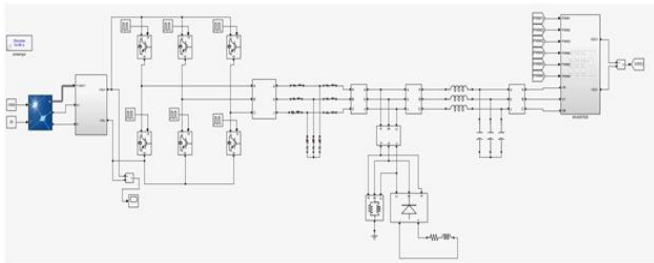


Fig. 6 Overall Simulink model of Solar Powered Grid

The irradiance of solar array has been varied to observe the power at those different irradiances and is shown in Fig 7. The maximum power 100kW is achieved at the standard irradiance level of 1000W/m². The output voltage from the boost converter is shown in Fig.8. As the irradiance levels are varying, the boost converter outputs are also changing accordingly. With the use of MPPT, the error with the reference voltage is reduced. The boost converter output is fed to the three phase inverter. It has also been observed that the voltage and current waveforms at three phase inverter before filter are not pure. Hence the LC filter is used for smoothening the output from the inverter. The output of the three phase inverter with the LC filter is shown in Fig. 9.

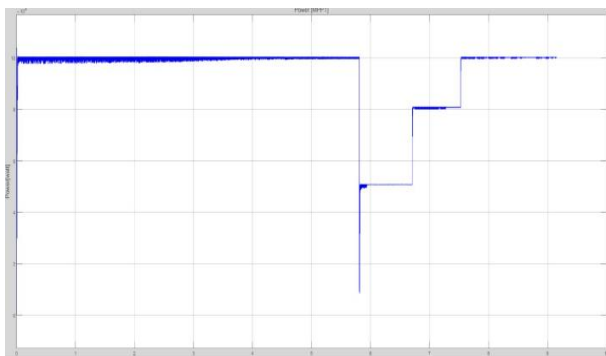


Fig 7 Solar Power at Different Irradiances

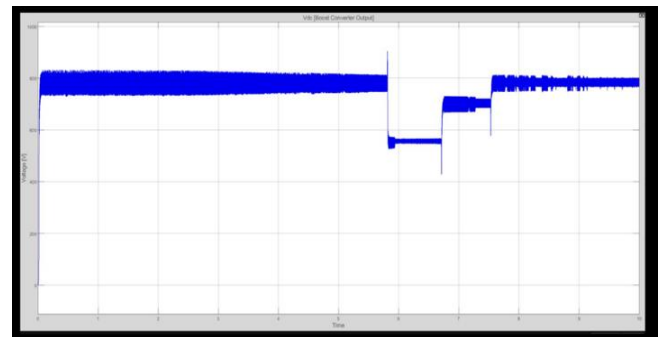


Fig 8- Voltage Waveform Boost converter

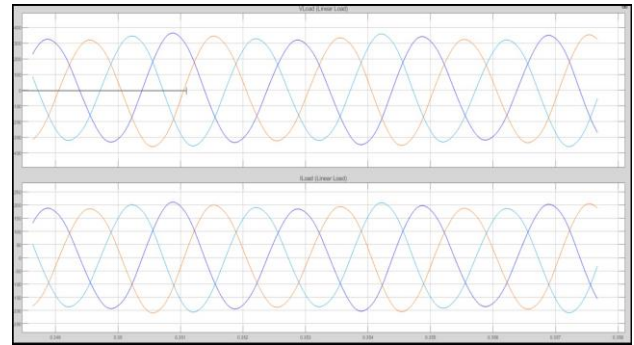


Fig 9- Voltage and Current Waveforms with filter

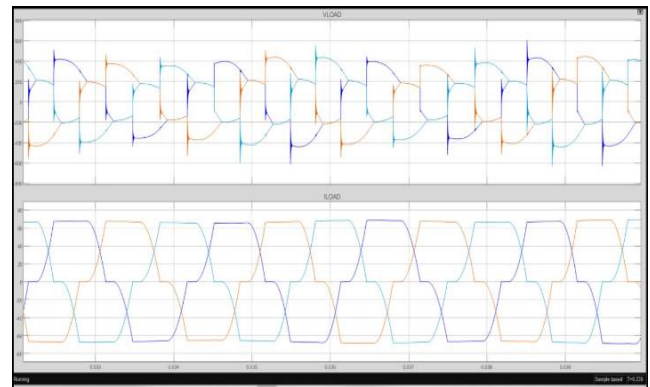


Fig 10- Load Voltage and Current without STATCOM

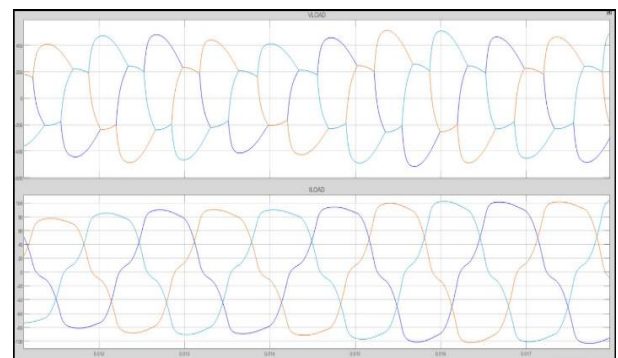


Fig 11- Load Voltage and Current with STATCOM

The filter is utilized to make the stepped three phase output waveform of the inverter into the shape of sine waveform. The comparative analysis of voltage and current waveforms with STATCOM and without STATCOM are done. The 3 phase load voltage and current without the addition of STATCOM and with STATCOM are shown in Fig. 10 and 11 respectively. The harmonics present in the load voltage

and current are reduced with the addition of STATCOM. This shows that the power quality of PV array based system is improved. More harmonics in the voltage and current cannot be fed to the load directly as it harms the loads.

V. CONCLUSION

The design and implementation of a standalone solar powered AC transmission systems for non-linear loads are discussed in detail. The power quality of the system under varying load conditions is improved by designing the D-STATCOM. The control algorithms for the solar system and D-STATCOM and its analysis are discussed in detail. The proposed model can be suggested for the standalone solar system and the proposed control is well suited to improve the power quality of the overall system. The overall performance of the system has been improved under different solar irradiance levels and varying load conditions.

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